

PROJECT REPORT

TEAM NAME: MurubbEEE

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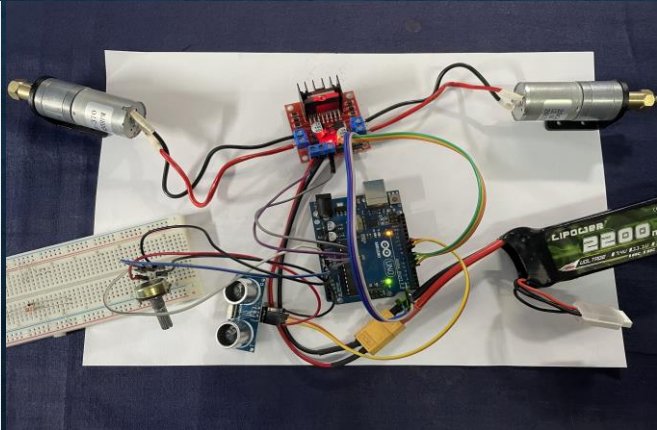
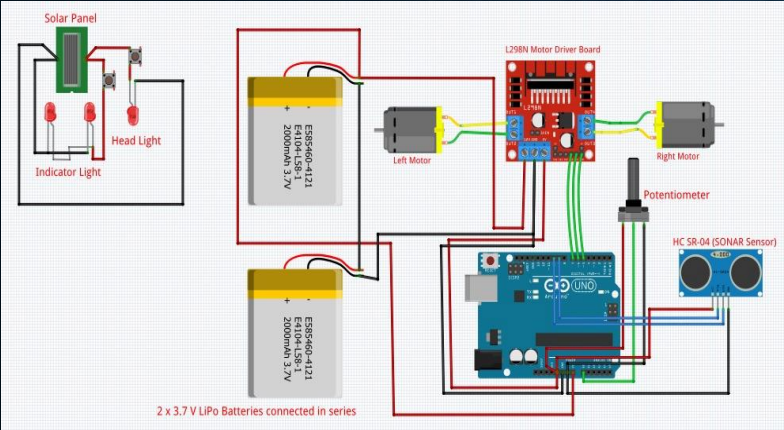
PROJECT: ENHANCEMENT OF A BATTERY POWERED RICKSHAW

We, Team MurubbEEE aimed to design a battery powered rickshaw in which there will be some features like: speed control of motor, obstacle detector (automatic breaking), indicator lamps and headlamps using solar panel. Using Fritzing , We have formulated a circuit where we have used potentiometer ,sonar sensor ,LED,solar panel ,motors , aurduino UNO and motor drivers . Features that have been processed so far:

- **Speed Control :**
 - For speed controlling , we have used potentiometer. Potentiometer is actually a variable resistor , by varying resistance we can control current which will also control the acceleration .
- **Obstacle Detection :**
 - For obstacle detection , we have used sonar sensor(HC SR-04) .
- **Solar-Powered Indicators and Headlights:**
 - We want solar panel to provide required power to the indicator and headlights .(All of them will be LED)

Code ,Circuit Design & Implementation

-Using Aurduino ,we have developed a code for operating the internal circuit efficiently .
We have also implemented a circuit circuit also which is given below :



Circuit Design & Implementation

Primarily selected components

Sl. No	Components	Sl. No.	Components
1	Aurduino – UNO R3	5	11V LiPo Battery
2	Motor driver- L298N	6	Motors
3	Potentiometer (10K or 50K)	7	Solar panel
4	Sonar sensor- HC SR-04	8	LED's